Issues and Resolutions to Open Problems in *Rational Rotations of Three Dimensional Minkowski Space*

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Abstract

This companion addresses N. J. Wildberger's revised Minkowski paper. It records the constructive formalisation of its finite algebraic assertions, clarifies null-axis spread and the inverse of a negative-norm ray, and separates causal character from arithmetic splitting. It answers the coefficient-product question in Problem 90 and an explicit projective boost–rotation factorization for Problem 92, including the half spread and spinor norm of the Thomas–Wigner rotation. The general ternary Clifford correspondence answers Problem 93. Finite-word spinor calculations give partial information for Problem 91, while the integral programme in Problem 94 remains open. Results with existing Lean certificates are distinguished from the additional coordinate proofs supplied in this companion.

1 Source, conventions and status

The source is the supplied 36-page revision of *Rational rotations of three dimensional Minkowski space: rotations, boosts and spinors* [1]. Its numbered assertion inventory ends at Theorem 89 and its research questions are Problems 90–94. Vectors are rows in the time-first convention

$$G = \operatorname{diag}(1,-1,-1), \quad B(v,u) = vGu^T, \quad Q(t,r,s) = t^2 - r^2 - s^2.$$

All general-field statements assume characteristic different from two. The signed cross product is exactly the source's

$$v \times u = (s_v r_u - r_v s_u,\ s_v t_u - t_v s_u,\ t_v r_u - r_v t_u).$$

The dihedron multiplication, norm and matrix map are

$$(x:v)(y:u) = (xy - B(v,u) : xu + yv - v \times u), \qquad N(x:v) = x^2 + Q(v), \tag{1}$$

$$\Phi(x:t,r,s) = \begin{pmatrix} x+s & r-t \\ r+t & x-s \end{pmatrix}, \qquad \det \Phi = N. \tag{2}$$

The product is multiplicative in this matrix order. The rotation invariant is $p(\rho) = (3 - \operatorname{tr}\rho)/4$. For a cap represented by $(1:v)$,

$$p = \frac{Q(v)}{1+Q(v)}, \qquad \chi = [1+Q(v)] = [1-p].$$

The repository's older rational Minkowski development uses the different form $\operatorname{diag}(1,1,-1)$. Its vectors and cross-product signs cannot be read as the coordinates above. The formal comparison uses an explicit time-first form owner and a vector sign conversion; all examples below use (2) directly.

Source item	Result of this companion
Definition 45 / Theorem 46	Extension to zero and nonzero null anchors, with an explicit non-null test direction.
Section 12.3	Exact inverse ray for both signs of the norm; distinction between representative rays and anchor-vector rays.
Problem 90	Necessity and sufficiency in characteristic different from two of $\delta = -\gamma^2$, by the coordinate proof below.
Problem 91	Finite-word determinant classes and kernel tests; no complete classification of generated subgroups or arithmetic subgroups.
Problem 92	Projective Wigner factor, its half spread and spinor norm, for two future pure boosts; factor order is specified.
Problem 93	General ternary projective Clifford correspondence.
Problem 94	Elementary primitive-representative reduction is described; the integral classification programme is not solved.

2 Null axes and the measurement of spread

Definition 45 defines spread for a cap with a non-null anchor, or a half turn. The prose after Theorem 46 also discusses parabolic rotations with $p = 0$. To attach its divided vector-spread measurement to these rotations, the definition and existence argument should explicitly include null anchors. The nondegenerate-perpendicular-plane argument in its printed proof alone does not establish that extension.

For a nonzero null vector $v = (t, r, s)$, a non-null perpendicular vector can be chosen without a square root. If $t \neq 0$, take

$$w = (0, s, -r), \qquad B(v, w) = 0, \quad Q(w) = -r^2 - s^2 = -t^2 \neq 0.$$

If $t = 0$, take $w = (1, 0, 0)$. The same latter vector handles the zero anchor. For a non-null anchor the existing finite perpendicular construction applies. Thus an admissible w exists in every case, over any field of characteristic different from two.

The proof of the formula does not require $Q(v) \neq 0$. For any defined cap and any supplied $w \perp v$ with $Q(w) \neq 0$,

$$w\rho = \frac{(1 - Q(v))w + 2(v \times w)}{1 + Q(v)}, \qquad s(w, w\rho) = 4p(1 - p).$$

Consequently a null cap has measured spread zero even when it is not the identity. One proposed editorial repair is to let Definition 45 use any cap anchor and insert the displayed null-anchor construction in Theorem 46. This strengthens the stated domain without altering its formula. Non-null test vectors remain essential: the denominator of vector spread is not defined on a null test vector.

3 Ray inverses and rational normalization

3.1 The sign-sensitive inverse

A ray identifies P with aP only for $a > 0$. A projective proportion allows every $a \neq 0$, so a non-null projective point contains two opposite rays. The exact dihedron inverse is

$$P^{-1} = \frac{\bar{P}}{N(P)}, \qquad \overline{(x : v)} = (x : -v). \tag{3}$$

This identity proves both inverse laws, since $P\bar{P} = \bar{P}P = N(P)1$. It also shows that inversion respects positive scaling: $(aP)^{-1} = a^{-1}P^{-1}$.

For $N(P) > 0$, the inverse ray is the ray of $\bar{P}$. For $N(P) < 0$, it is the ray of $-\bar{P}$. Using the conjugate alone in that second case gives a product in the ray of -1, not the identity ray. Projectivization hides this sign, but positive rays retain it. The source's full ray-group assertion is correct with (3); making the representative of the inverse explicit avoids this ambiguity.

For the source's negative-norm example,

$$P = (2 : 1, 2, 3), \quad N(P) = -8, \quad \Phi(P) = \begin{pmatrix} 5 & 1 \\ 3 & -1 \end{pmatrix},$$

the inverse in the source coordinates is

$$P^{-1} = (-\tfrac{1}{4} : \tfrac{1}{8}, \tfrac{1}{4}, \tfrac{3}{8}), \qquad \Phi(P^{-1}) = \begin{pmatrix} 1/8 & 1/8 \\ 3/8 & -5/8 \end{pmatrix}.$$

Both matrix products are the identity; its norm is $-1/8$. This exact example and its sign conversion into the retained Clifford coordinates are checked by the facade consumer.

3.2 Two different sorts of rays

Opposite dihedron rays P and $-P$ represent the same rotation. Faithfulness of the projective action says that its rotation fibers consist of exactly these two rays, and the kernel is the scalar rays of 1 and -1. By contrast, opposite nonzero *anchor-vector* rays v and $-v$ represent inverse cap rotations. This distinction holds for timelike, spacelike and null anchors. It is the anchor-vector distinction which records the source's sense of orientation.

The norm-one representative question is separate again. A rational ray can be rescaled to norm one precisely when $N(P)$ is a nonzero rational square. If it exists, the positive rescaling and the unit representative are unique within that ray. Negative norm makes normalization to $+1$ impossible; positive norm alone is insufficient over $\mathbb{Q}$. The introductory boost $(3 : 0, 0, 1)$ has norm eight and class [2], so has no rational determinant-one representative. Its projective matrix nevertheless gives the exact rational boost

$$\begin{pmatrix} 5/4 & 3/4 & 0 \\ 3/4 & 5/4 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Hence rational dihedron rays cover the whole group twice, whereas rational determinant-one representatives cover only $\ker \chi$.

4 Causal type and arithmetic splitting

Theorem 89 uses the discriminant

$$\Delta = -16p(1-p).$$

For a prescribed non-null axis v, write its cap anchor as λv, where $\lambda \neq 0$. The existence equation is $\lambda^2 Q(v) = p/(1-p)$, and therefore

$$\Delta = \big(4\lambda(1-p)\big)^2(-Q(v)).$$

Thus the arithmetic splitting condition is precisely that $-Q(v)$ be a square in the ground field. It is not just that the axis is spacelike. The retained square witness constructs both nontrivial eigenvalues and actual nonzero eigenvectors of the original three-by-three matrix.

The distinction is already correct in the revised source; it is worth making its exact examples explicit. The introductory boost has $p = -1/8$ and nontrivial eigenvalues $2, 1/2$. The rational axis $(1,2,3)$ has $Q = -12$, and is spacelike, but 12 has no rational square root. The formal example with anchor $(1/4)(1,2,3)$ has $p = -3$, time entry $15/2 > 0$, and no nontrivial rational eigenvalue. It is causally boost-type while arithmetically nonsplit over $\mathbb{Q}$.

The non-orthochronous example $(2 : 1,2,3)$ has $p = 3/2$, time entry $-9/4$, and characteristic polynomial

$$(X-1)\big((X+2)^2 - 3\big).$$

Over a field supplied with $r^2 = 3$, its other roots are $-2 \pm r$. They are not rational. Causal sign ranges, determinant square classes and ground-field eigenvalue splitting are therefore three separate pieces of information; none substitutes for the other.

5 Problem 90: associative products

The revised question asks about

$$v \star u = \frac{v + u + \gamma(v \times u)}{1 + \delta B(v,u)}, \qquad \gamma \neq 0.$$

Theorem 5.1 (Lorentz coefficient classification)**.** *This product is associative wherever all four products in the two bracketings are defined if and only if $\delta = -\gamma^2$.*

Proof. Take $e_0 = (1,0,0)$ and $e_1 = (0,1,0)$. For any supplied nonzero a with $D = 1 + \delta a^2 \neq 0$, the Lorentz coordinate identities

$$B(e_0,e_0) = 1, \quad B(e_0,e_1) = 0, \quad e_0 \times e_1 = (0,0,1), \quad e_0 \times (0,0,1) = -e_1$$

give, for the triple (ae_0, ae_0, e_1), the same two middle coordinates as in the Euclidean probe:

$$\big((ae_0 \star ae_0) \star e_1\big)_r = 1, \qquad \big(ae_0 \star (ae_0 \star e_1)\big)_r = \frac{1 - \gamma^2 a^2}{1 + \delta a^2}.$$

All four denominators are nonzero. Equality forces $(\delta + \gamma^2)a^2 = 0$. Try $a = 1$. If its denominator is zero, then $\delta = -1$; when $\gamma^2 = 1$ the required relation is already true, and otherwise the nonzero probe $a = \gamma$ has denominator $1 - \gamma^2 \neq 0$. These explicit alternatives prove necessity in every stated characteristic.

For sufficiency the dilation $v \mapsto \gamma v$ identifies the product with $(v + u + v \times u)/(1 - B(v,u))$, the opposite dihedron affine product. The associative homogeneous matrix product (1) proves equality on every common affine domain. The dilation is invertible and preserves precisely the displayed denominators. □

This answers the revised question over every field of characteristic different from two, including its primary rational case. The rational coefficient classification already has a Euclidean Lean certificate. The Lorentz necessity calculation above is a proof supplied in this paper, not a newly added standalone Lean theorem.

6 Problem 91: finite spinor calculations and remaining subgroup theory

The determinant class of a supplied rational invertible matrix is computable exactly. A nonzero rational c has the square-class signature consisting of its sign and the primes occurring to odd exponent in its reduced numerator and denominator. Sign and prime parities add modulo two under multiplication. The sign coordinate is essential in the Lorentz case: χ takes values in all rational nonzero square classes, not just the positive ones.

Given finitely many generator matrices, every supplied finite signed word can therefore be evaluated by exact two-by-two products. Multiplicativity of the determinant gives its spinor class, and a word is in $\ker \chi$ precisely when its determinant is a rational square. Comparing the class of one word times the inverse of another gives an exact kernel-coset test. The finitely many generator sign/parity vectors span a finite binary space, whose vectors and generator words can be enumerated. These calculations describe the spinor image detected by the generators. They do not determine the internal relations of the generated subgroup.

In particular, if the generators are already in $\ker \chi$, every word has trivial spinor norm, but it may still be a different group element. There is no internal subgroup classification hidden in this test. The repository's complete generated-subgroup certificate is for the Euclidean positive-norm case; its exact rational matrix and square-class laws supply the ingredients here, but no separate Lorentz subgroup-classification certificate is asserted.

The complete Euclidean finite-order table in the other companion cannot be silently transferred to the full Lorentz group. For example, a Lorentz half turn has $\chi = [Q(v)]$, which may be negative. The norm-square test answers the question for any supplied matrix or order certificate, but a complete classification of finite-order spinor values and natural arithmetic subgroups is not supplied here. These parts of Problem 91 remain open.

7 Problem 92: a projective Thomas–Wigner factor

We specify the convention first: factor the product of two future pure boosts as a future pure boost followed by a spatial rotation, in the source's matrix order. Changing this convention changes the factor one calls the Wigner rotation. No angle or square-root normalization is needed for the following factorization.

7.1 The factorization and its invariants

Let the boost anchors be

$$v = (0, r_1, s_1), \qquad u = (0, r_2, s_2), \qquad r_i^2 + s_i^2 < 1.$$

These inequalities select the future pure-boost representatives

$$A = \Phi(1 : v), \qquad B = \Phi(1 : u),$$

which are symmetric with positive determinant and positive trace. Define

$$d = 1 - B(v,u) = 1 + r_1 r_2 + s_1 s_2, \qquad k = r_1 s_2 - s_1 r_2, \qquad H = d^2 + k^2. \tag{4}$$

Let $r = r_1 + r_2$, $s = s_1 + s_2$. The product dihedron is $P = (d : k, r, s)$, so $AB = \Phi(P)$.

Theorem 7.1 (Projective boost–rotation decomposition)**.** *Over $\mathbb{Q}$, or any ordered field, the quantities in* (4) *satisfy $d > 0$, $H > 0$. Put*

$$R = \begin{pmatrix} d & -k \\ k & d \end{pmatrix}, \qquad S = \begin{pmatrix} H + sd - rk & rd + sk \\ rd + sk & H - sd + rk \end{pmatrix}.$$

Then

$$AB = H^{-1} SR. \tag{5}$$

The projective class of S is a future pure boost and that of R is a spatial rotation. The corresponding Wigner rotation ω has

$$p(\omega) = \frac{k^2}{d^2 + k^2}, \qquad \chi(\omega) = [d^2 + k^2]. \tag{6}$$

This spatial-rotation factor is unique among factorizations in this order with a future pure-boost factor.

Proof. Write $a = r_1 r_2 + s_1 s_2$ and $b_i = r_i^2 + s_i^2 < 1$. The identity $a^2 + k^2 = b_1 b_2$ gives $a^2 < 1$, hence $a > -1$ and $d = 1 + a > 0$. Thus $H > 0$. Direct multiplication gives

$$\Phi(P)\,\mathrm{adj}(R) = S, \qquad R\,\mathrm{adj}(R) = HI,$$

and therefore (5). The matrix S is symmetric and

$$\det S = (\det A)(\det B)H = (1 - b_1)(1 - b_2)H > 0, \qquad \mathrm{tr}\, S = 2H > 0.$$

Its dihedron has coordinates $(H : 0, rd + sk, sd - rk)$. After scaling by $H^{-1} > 0$, its scalar coordinate is one, its time anchor coordinate is zero, and its norm is positive. Thus its spatial anchor has squared length less than one, exactly the future pure-boost condition. The matrix R has dihedron coordinates $(d : k, 0, 0)$, so it acts as a spatial rotation fixing the time axis. The norm and half-spread identities give (6).

For uniqueness, any non-null spatial-rotation representative has the form $R' = \left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$. Symmetry of $\Phi(P)\,\mathrm{adj}(R')$, required of its boost factor, is equivalent to $ka = db$. Since $d \neq 0$, nonzeroness of R' forces $a \neq 0$, and its projective class is exactly that of R. □

This gives a closed formula for the *rotation factor*, rather than merely detecting a time component in the product anchor. It stays over the ground field: normalization of either two-by-two factor to determinant one may require a square root, but the projective factors do not.

For comparison, the half spread and spinor norm of the *whole product* are

$$p(\rho(v)\rho(u)) = 1 - \frac{d^2}{(1 + Q(v))(1 + Q(u))}, \qquad \chi(\rho(v)\rho(u)) = [(1 + Q(v))(1 + Q(u))]. \tag{7}$$

These are generally different from (6). They follow directly from the homogeneous scalar coordinate and multiplicative norm; calling them the Wigner rotation's invariants would conflate the boost with the rotation factor.

7.2 The perpendicular-boost example

For Example 69 take

$$v = (0, 0, 1/3), \qquad u = (0, 1/3, 0).$$

Then $d = 1$, $k = -1/9$, and $H = 82/81$. Reversing the boosts changes k to $-k$ and inverts the Wigner rotation. The projective rotation representative is $(1 : -1/9, 0, 0)$, and

$$p(\omega) = \frac{1}{82}, \qquad \chi(\omega) = [82].$$

The whole product instead has $p = -17/64$ and trivial spinor norm, since both boost norms are $8/9$. A convenient rational factorization is

$$\Phi(P) = \begin{pmatrix} 4/3 & 4/9 \\ 2/9 & 2/3 \end{pmatrix} = \frac{81}{82} \begin{pmatrix} 112/81 & 8/27 \\ 8/27 & 52/81 \end{pmatrix} \begin{pmatrix} 1 & 1/9 \\ -1/9 & 1 \end{pmatrix}.$$

The residual future pure boost has anchor $(0, 12/41, 15/41)$. Its spinor class is also $[82]$, so the two nontrivial factor classes cancel in the product. This is an exact rational example of why the product's spinor norm alone does not determine that of the Wigner rotation.

The projective product and Example 69 are already Lean-checked. The factorization theorem, uniqueness argument and its Wigner formulas are additional coordinate proofs in this paper. They have not been added as standalone Lean declarations, and the earlier full audit should not be represented as certifying those new theorem statements.

8 Problem 93: the general ternary correspondence

For a symmetric nonsingular matrix M, put

$$B_M(u, v) = uMv^T, \quad Q_M(u) = B_M(u, u), \quad d = \det M, \quad C_M(u, v) = (u \times_{\text{coord}} v)\,\text{adj}(M).$$

The general even Clifford product is

$$(x : u)(y : v) = (xy - dB_M(u, v) : xv + yu - C_M(u, v)), \qquad N_M(x : u) = x^2 + dQ_M(u).$$

Its conjugation, associative product and multiplicative norm give the action

$$(0 : w) \longmapsto N_M(P)^{-1} \bar{P}(0 : w)P.$$

The retained constructive two-reflection theorem produces a preimage of every supplied member of $\text{SO}(M, F)$, and equality of actions is equivalent to an explicit nonzero scalar relation between the original non-null representatives. This is the projective correspondence requested in Problem 93, with the determinant factors and ownership of M retained. The proof and affine formula are stated in the Euclidean companion [2]; the same subject-owned Lean provider serves both papers.

For the present G, $d = 1$ and the generic form-relative cross is the *negative* of the paper's signed Lorentz cross. Negating the vector coordinate of the generic Clifford representative identifies its product with (1). This explicit conversion explains why one may share the abstract theorem without changing the source's signs.

The null quadric has a separate exact description. A nonzero null dihedron maps to a nonzero singular two-by-two matrix. Testing its four possible nonzero pivot entries constructs a nonzero column ξ and row η with $\Phi(P) = \xi\eta$. Its only factorization ambiguity is

$$(\xi, \eta) \longmapsto (a\xi, a^{-1}\eta), \qquad a \neq 0.$$

Holding either projective spinor fixed gives a ruling; distinct rulings in one family are disjoint, while a column ruling and a row ruling meet exactly in their outer-product point. The finite-field lists have exactly $(q+1)^2$ such projective null points and $q(q^2-1)$ non-null rotation points. These counts and all four pivot charts are part of the checked Minkowski facade surface, not consequences of an assumed completed quadric.

9 Problem 94: the integral programme

Every supplied rational non-null proportion can be reduced to an integral primitive four-vector: clear the four denominators, divide by their common integer gcd, and choose the sign of the first nonzero coordinate. This gives a unique representative for its rational projective point. Two primitive integer four-vectors representing the same rational point differ only by sign: a reduced rational scale's denominator must divide all four input coordinates, and its numerator must divide all four output coordinates. Both gcds are one.

Products of integral dihedrons are integral by (1), and their norms multiply. Thus one finite composition is performed by multiplying the four coordinates and then removing the resulting common gcd and sign. These elementary observations specify the arithmetic operation, but do not give a classification of its primitive norm fibers or integral subgroups.

The matrix lattice deserves an explicit qualification. An integral matrix $\left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$ has integral dihedron coordinates exactly when

$$a \equiv e \pmod 2, \qquad b \equiv c \pmod 2,$$

because its inverse coordinate map is

$$x = (a+e)/2, \quad s = (a-e)/2, \quad r = (b+c)/2, \quad t = (c-b)/2.$$

Primitive four-coordinates do not necessarily give a primitive matrix: $(1:0,0,3)$ is primitive and non-null, but its matrix $\operatorname{diag}(4,-2)$ has gcd two. Conversely, a primitive matrix need not lie in this parity lattice before rescaling. An integral theory must specify which primitive object it normalizes; rational projective equality does not by itself settle that lattice question.

These are introductory arithmetic reductions proved directly here. No standalone Lean implementation of the primitive-normalization procedure, full arithmetic orbit classification, presentation of integral subgroups or local-to-global norm theorem is claimed. Problem 94 remains open as that larger programme.

10 Formal evidence, finite boundaries and open work

The facade is `formalisations/MinkowskiRotations/MinkowskiRotations.lean`, and its claim ledger is `formalisations/MinkowskiRotations/coverage.md`. It covers the finite algebraic assertions numbered 1–89 and the unnumbered constructions, with direct source-coordinate facade consumers. Entry points include the following owners.

Checked result	Mathematical owner
Source vector and anchor signs	`Purelib/Geometry/Quadratic/Ternary/Lorentz.lean`.
Dihedron/matrix dictionary	`Purelib/Algebra/Clifford/Lorentz/Matrix.lean`; exact product, determinant and two coordinate inverses.

Checked result	Mathematical owner
Composition and velocities	`Purelib/Geometry/Rotations/Ternary/Lorentz/Composition.lean`; the partial anchor law and exact Einstein velocity identity.
Ray inversion and fibers	`Purelib/Algebra/Clifford/Ternary/Ray.lean`; `Purelib/Geometry/Rotations/Ternary/Clifford/RayCover.lean`.
Prescribed axes and splitting	`Purelib/Geometry/Rotations/Ternary/Lorentz/Prescribed.lean`; `Purelib/Geometry/Rotations/Ternary/Lorentz/Splitting.lean`.
Null spinors and rulings	`Purelib/LinearAlgebra/Coordinates2/RankOne.lean`; the original Lorentz matrix recovery and finite null-point enumeration.
General ternary correspondence	`Purelib/Geometry/Rotations/Ternary/Clifford.lean`; `projectiveCliffordCorrespondence`.

The mathematical tree at commit `a6219f044` passed all twelve completion gates on 11 October 2026, including the warning-fatal full build and lint, full configured-script axiom audit and independent nanoda pass. Build phases used 64 Lean threads. The normal audit checked 136,660 project declarations and 1,536 selected canonical declarations with no allowed axioms; nanoda accepted all 42 compatible module sets. This is the selected mathematical proof closure, not a strict audit of every internal meta-provider helper. The Lorentz proof for Problem 90, the Wigner factorization for Problem 92 and the integral reductions for Problem 94 are printed proofs introduced here, outside that existing Lean declaration inventory.

The infinite-index assertion of Corollary 86 has retained finite content: for every supplied n, construct n pairwise distinct rational square classes, preimages with those norms and pairwise kernel-separated rotations. The source norm witness

$$\left(\frac{c+1}{2} : 0, \frac{c-1}{2}, 0\right) \quad \text{has norm } c$$

works for every requested nonzero rational c. This supplies arbitrary finite coset packs without forming a completed quotient carrier. Real square roots, topological connected components, completed Lie groups and analytic angle/rapidity dictionaries remain outside the formal scope. Their absence is a constructive boundary, not a claimed defect in the source.

Problem 90 has the general-field answer above. Problem 92 has the displayed projective decomposition for the stated pure-boost convention, and Problem 93 has the general constructive correspondence. Problem 91 still asks for internal subgroup and arithmetic classifications not obtained by finite spinor tests. Problem 94 still asks for an integral theory beyond primitive-coordinate reduction. Formalising the additional printed proofs would be separate work; these companions do not imply that it has already been done.

References

[1] N. J. Wildberger, *Rational rotations of three dimensional Minkowski space: rotations, boosts and spinors*, supplied revision `RelativisticsRationalRotations3.pdf`, 36 pages. Source SHA-256: `8c397b21f726bf0a8c89b8c56bf84edf5d2586bcac552c0f10134d3c9fff46d7`.

[2] *Issues and Resolutions to Open Problems in Rational Rotations of Three Dimensional Euclidean Space*, companion in `formalisations/RationalRotations/IssuesAndOpenProblems.pdf`, 12 October 2026.